WEEK 5 · LECTURE L3 · UNIT II

Going Live: Diffusion and the Digital Divide

How new HR technology spreads through a workforce, why it stalls, and who it leaves behind.

R K Singh · Senior Professor, Department of Commerce · Delhi School of Economics, University of Delhi

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AFTER THIS LECTURE YOU SHOULD BE ABLE TO

What this lecture prepares you for

1. Explain how new HR technology spreads, using Rogers’ five perceived attributes and adopter categories.
2. Identify who is left behind, using van Dijk’s four kinds of access and Hargittai’s second level divide.
3. Distinguish an expected post go live dip (the J curve) from failure.
4. Use champions and segmented usage data to bring late adopters along, and state what usage data cannot show.

ONE SOURCE

The same theories as the in class lecture and the course hub, without the simulation data used in class. Course outcomes: CLO 2 and CLO 5.

COMMIT BEFORE THEORY

Commit to an answer first

Three months after a new HR app goes live, 30% of staff use it. The CHRO calls the launch a failure. What is the best response?

A

Agree: 30% after three months means the app has failed

B

It depends on who the 30% are and how fast use is spreading, not on the number alone

C

Disagree: 30% is the industry average

D

Agree, and replace the vendor

EXAM LENS

The answer is B. Diffusion theory and the digital divide explain why.

ROGERS (2003)

Five perceived attributes that speed adoption

1 · Relative advantage

Better than what it replaces, as users see it.

2 · Compatibility

Fits existing values, habits and needs.

3 · Complexity

Easy to understand and use; lower is faster.

4 · Trialability

Can be tried in small steps first.

5 · Observability

Results visible to others.

EXAM LENS

Perceived attributes explain much of the variation in how fast innovations are adopted. The word that matters is perceived: an app is “advantageous” only if its users think so.

ROGERS (2003)

Adopter categories: the S curve and the bell curve

Category	Share of adopters	Who they are
Innovators	2.5%	Venturesome; tolerate failure
Early adopters	13.5%	Respected, watched by others: the natural champions network
Early majority	34.0%	Deliberate; adopt just before the average
Late majority	34.0%	Skeptical; adopt under pressure or when risk falls
Laggards	16.0%	Last to adopt; often anchored in the past

EXAM LENS

Categories are defined by standard deviations from mean adoption time; cumulative adoption traces an S curve. A caution: late adopters in HR are often those with least access, not least will. The label laggard can hide exclusion.

VAN DIJK (2005; 2020)

Four kinds of access, in sequence

1 · Motivational

Wanting to use the technology at all.

2 · Material

Having a device, data and a connection.

3 · Skills

Operating it, finding information, using it strategically.

4 · Usage

Using it often, and for things that matter.

EXAM LENS

Giving people a device solves the second step only; motivation, skills and meaningful use remain.

HARGITTAI (2002)

The second level digital divide

First level

Who has access to a device and the internet.

Second level

Among those online, differences in skills: finding, judging and using information.

What it means for HR rollouts

An app that assumes English, typing and confident navigation excludes users who are online but not fluent.

EXAM LENS

Usage data that counts logins hides the second level divide; segment by skill and language. Week 2’s third level, outcomes (van Deursen and Helsper, 2015), asks whether the benefit actually arrives.

BRYNJOLFSSON, ROCK AND SYVERSON (2021)

Why things look worse after go live: the J curve

The idea

General purpose technologies need complementary intangible investment (new processes, skills, organization) before gains appear, so measured productivity first dips, then rises.

Expected

A dip while people and processes adjust: more helpdesk tickets, slower transactions, parallel running.

Warning sign

A dip that does not turn after the adjustment period.

COURSE FRAMING

A temporary dip is expected; the test is whether it turns. Apply the idea to an HR rollout as course framing: the paper is about the economy, not a single app.

BRINGING THE LATE ADOPTERS ALONG

Champions, compatibility and segmented evidence

Champions

Choose early adopters respected on the floor, not only enthusiasts in head office; give them time and recognition.

Compatibility and complexity

Vernacular voice and video, fewer taps, no typing; assisted options for those without skills or devices.

Observability and trialability

Let staff see a peer complete a task; offer low risk first tasks.

Evidence

Segment adoption by role, location, device and language; measure meaningful use, not logins.

Course framing, built on Rogers (2003) and van Dijk (2020).

APPLY

An HR app for a dairy cooperative

Teaching case (Course framing)

A Gujarat dairy cooperative launches a self service app for 9,000 staff, from plant workers to collection center agents. After four months, 25% use it, almost all office staff and shift supervisors. Plant workers have smartphones but say the app is in English and asks them to type. Tickets to HR have risen.

Work it through

1. Which of Rogers' attributes is the app failing on?
2. Which of van Dijk's access stages blocks plant workers?
3. Is the rise in tickets a reason to stop, using the J curve idea?
4. Who should be your champions, and why?

COMMON CONFUSIONS

Tempting readings, and better ones

A tempting reading	A better reading
Everyone has a phone, so access is solved	Material access is one of four; skills and meaningful use come after it
Late adopters are resistant	Many late adopters in HR lack access or skill
A dip after go live means failure	A temporary dip is expected; the test is whether it turns
Logins measure adoption	Logins hide skill, purpose and quality of use

COMPARE AND EVALUATE

Lenses on going live

Lens	Question it answers	Watch out for
Rogers: attributes	Why does adoption speed vary?	Perceptions, not features
Rogers: categories	Who adopts when?	Labels can hide exclusion
van Dijk (2005; 2020)	Which kind of access is missing?	Sequence varies by person *
Hargittai (2002)	Who is online but not fluent?	Skills are hard to measure
J curve (2021)	Is the dip expected?	An economy wide idea applied to one rollout *

* Course framing.

PRACTICE FOR THE END TERM EXAMINATION

Questions to rehearse

Six months after launch, a Kolkata bank’s HR app is used by 70% of head office staff and 20% of branch staff. Using diffusion of innovations and the digital divide, explain the pattern and propose how to bring branch staff along. Note what usage data cannot tell you. **20 marks**

A strong answer: applies Rogers’ attributes and adopter categories; uses van Dijk’s four kinds of access and Hargittai’s second level divide; distinguishes a J curve dip from failure; notes that login counts cannot show skill, purpose or quality of use.

“Giving every employee a smartphone solves the digital divide in HR.” Critically evaluate. **20 marks**

A strong answer: uses van Dijk’s sequence and Hargittai’s second level; adds the outcomes level; proposes design and capability responses.

Explain Rogers’ adopter categories and one caution in applying them to an HR rollout. **10 marks**

A strong answer: gives the five categories and shares; warns that late adoption may reflect access, not will.

CARRY THESE INTO THE EXAMINATION

Four things to remember

1. **Adoption speed follows perceived attributes.**
   Advantage, compatibility, complexity, trialability, observability.

2. **Access comes in four kinds, in sequence.**
   A device solves only the second.

3. **A dip is expected; a dip that does not turn is the warning.**
   Judge trajectories, not snapshots.

4. **Late is not always resistant.**
   Segment usage data and design for the least connected.

SET READINGS · FROM THE LESSON PLAN

Read for this week

- Scully-Russ, E., & Torraco, R. J. (2020), pp. 83 to 84.
- Case: Infosys Lex; NASSCOM FutureSkills (latest).

NEXT WEEK

- Bondarouk, Parry and Furtmueller (2017), adoption and consequences; Venkatesh et al. (2003); Manuti and de Palma (2018), Chapters 4 and 5.

> **NOTE**
> The other works named in this lecture are theory sources for reference, not set readings.

THEORY SOURCES · FOR REFERENCE

Sources for this lecture

- Brynjolfsson, E., Rock, D., & Syverson, C. (2021). The productivity J curve: How intangibles complement general purpose technologies. *American Economic Journal: Macroeconomics, 13*(1), 333 to 372. https://doi.org/10.1257/mac.20180386
- Hargittai, E. (2002). Second level digital divide: Differences in people's online skills. *First Monday, 7*(4). https://doi.org/10.5210/fm.v7i4.942
- Rogers, E. M. (2003). *Diffusion of innovations* (5th ed.). Free Press.
- van Deursen, A. J. A. M., & Helsper, E. J. (2015). The third level digital divide. In *Communication and information technologies annual* (Vol. 10, pp. 29 to 52). Emerald.
- van Dijk, J. A. G. M. (2005). *The deepening divide*. Sage.
- van Dijk, J. A. G. M. (2020). *The digital divide*. Polity.